

Numerical Study of the Effect of Reynolds Number on Mixing Efficiency in Inertial Microfluidic Mixers

ME 224: Fundamentals of Microscale Flow

Submitted by

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Abstract

Microscale mixing is a fundamental process in microfluidic systems used in biomedical diagnostics, chemical microreactors, and lab-on-chip technologies. However, due to the small dimensions of microchannels, fluid flow typically occurs at low Reynolds numbers where turbulence is absent and mixing is largely diffusion-limited.

This study presents a numerical investigation of the effect of Reynolds number on mixing efficiency in different inertial microfluidic mixer geometries. Four micromixer designs—rectangular ridges, curly baffles, fin-shaped structures, and river-shaped channels—were modeled and simulated using COMSOL Multiphysics.

The simulations were performed for water with Reynolds numbers ranging from 0 to 300, while tracking the transport of a passive scalar concentration to quantify mixing efficiency. The study analyzes the combined effects of diffusion, convective transport, and vortex formation on the mixing process.

Results show that micromixer geometry strongly influences mixing performance, with different designs exhibiting distinct mixing efficiency trends with increasing Reynolds number. Among the investigated designs, the curly baffle micromixer achieved high mixing efficiency at moderate Reynolds numbers due to curvature-induced centrifugal effects and secondary flows.

The results highlight the importance of channel geometry in enhancing mixing in inertial microfluidic systems and provide insights for designing efficient micromixers for microfluidic applications.

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1 Introduction

What is Microscale Mixing?

Microscale mixing refers to the process of combining two or more fluid streams inside channels with dimensions typically between 10–500 μm .

This is an important phenomenon considering the fact that highly efficient mixing would enable better reactions, homogeneity in fluids and better efficiency for devices that use mixing of microfluids.

This process is achieved by devices called Micromixers. There are different types of micromixers with various types of obstacle designs that serve as the main factors affecting the mixing efficiency of flow.

2 Literature Review

Recent advancements in inertial micromixers

Recent research on inertial micromixers indicates that mixing at moderate Reynolds numbers depends primarily on vortex formation and secondary flows.

- Studies on Y-channel designs show critical Reynolds numbers where flow transitions from symmetric to asymmetric, sharply increasing efficiency.
- In curved microchannels, Dean vortices provide consistent, high-performance mixing over different regimes.
- Furthermore, fin-shaped and curly baffles enhance inertial effects via curvature-induced chaotic advection.

3 Motivation and Objective

- **The need for efficient mixing in microfluids:** Widely used in Biomedical diagnostics, Drug delivery systems, Chemical microreactor etc.
- **Limitations:** Turbulence being absent in microfluids would mean mixing is diffusion limited.
- **Potential of Inertial Microfluids:** Recent research has shown that inertial microfluids can enhance mixing, thus helping our project to delve deeper into the factors that would affect inertial fluid mixing

Primary Objective: To numerically investigate the effect of Reynolds number on mixing efficiency in different inertial microfluidic mixer geometries.

Specific objectives:

- To design and simulate multiple micromixer geometries
- To compare micromixer designs based on mixing efficiency and mixing length.
- Identify optimal design and operating conditions.

4 Methodology

Methodology involves designing different micromixers with two inlets, where the same inertial fluid with different concentration is passed for them to then undergo mixing in the long channel. The variation of mixing efficiency with Reynolds number (Re) of the inlet flow is plotted (both inlet fluids will have the same Re) for different micromixer designs.

We use COMSOL Multiphysics software for the designing and simulation of different designs.

5 Theory

Flow Analysis

The fluid used for simulation is water. We are considering a Re range of $0 < Re < 300$.

In microfluids the Re is generally small since the hydraulic diameter of flow channel is in micro ranges, Hence it is almost impossible to achieve turbulence in microscale flow. This Re range ($Re > 0$ but not too big) would imply that the flow is inertial dominant with small laminar properties. For all intends and purposes, we can consider the flow to be inertial. Since turbulence is not achievable, the geometric properties of the micromixer will solely determine the factors that would affect the mixing.

Concentration of Mixing Fluid

The factor we are using as a measure of mixing index is the concentrations of fluids at inlet.

Concentration ‘c’, here is essentially a passive species that don’t change the fluids inherent hydrostatic properties since we require the same fluid at both the inlets. We consider $c=1$ and $c=0.5$ at inlets 1 and 2 respectively, which then undergoes mixing in the micromixer to ideally provide an outlet fluid of concentration 0.75 (or 0.5).

$c=1$ implies dye at inlet 1 and this enables us to track the flow and see how the fluid layers overlap and stretch at curves, edges and obstacles to enhance mixing.

As ‘c’ at outlet approaches 0.75, Mixing index/efficiency approaches unity.

Reynolds Number, Re

$$Re = \frac{\rho u d_h}{\mu}$$

This is the component of the experiment that is varying, corresponding to which mixing efficiency is calculated. Both the inlet fluids would have the same Re, i.e. simultaneous variation. Velocity is being varied in order to vary the Re. Factors affecting mixing are purely influenced by the

micromixer geometry and hence, we have done simulations for different types of micromixer channel designs.

The dominant factors affecting are:

- Diffusion
- Convective transport
- Vortexes

Diffusion

Random molecular motion causes overlapping and transfer of fluid particles, that is particles moves from region of higher concentration to region of lower concentrations.

$$J = -D\nabla c$$

where, J = Diffusion flux

D= Diffusion constant

c= concentration

But diffusion needs time to perform,

$$t_d = \frac{L^2}{D}$$

Where, L = length of flow.

Convection/Advection

This is the transport due to ‘bulk fluid movement’. In essence, when fluid moves everything inside it moves. Hence fluid velocity is the underlying factor.

$$u \cdot \nabla c$$

Vortex

It is a rotating region of fluid where the fluid circulated locally. Many such regions are present in the micromixer. Formed when fluid layers move at different speeds and directions, creating instability or rotation. Here there is direct mixing of fluid layers. Interface stretching and increase

in surface area of exposure, all thereby increasing mixing of fluids. Vortex contribution is contained in Governing Equations.

Mixing Strength : Vortex > Convection > Diffusion

Peclet Number, Pe

It is a number that compares convection to diffusion,

$$Pe = \frac{uL}{D}$$

In microfluids, $Pe \gg \gg 1$. Therefore, we conclude diffusion is very weak in microfluids. Also, as convection increases diffusion decreases.

Governing Equations

- | | |
|----------------------------------|---|
| 1. Continuity Equation | $\nabla \cdot \mathbf{u} = 0$ |
| 2. Navier Stokes Equation | $\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u}$ |
| 3. Convection-Diffusion Equation | $\mathbf{u} \cdot \nabla c = D \nabla^2 c$ |

Mixing Efficiency

The mixing efficiency or mixing index (MI) can be calculated by the expression,

$$MI = 1 - \sqrt{\frac{\int_A (c - \bar{c})^2 dA}{\int_A (c_0 - \bar{c})^2 dA}}$$

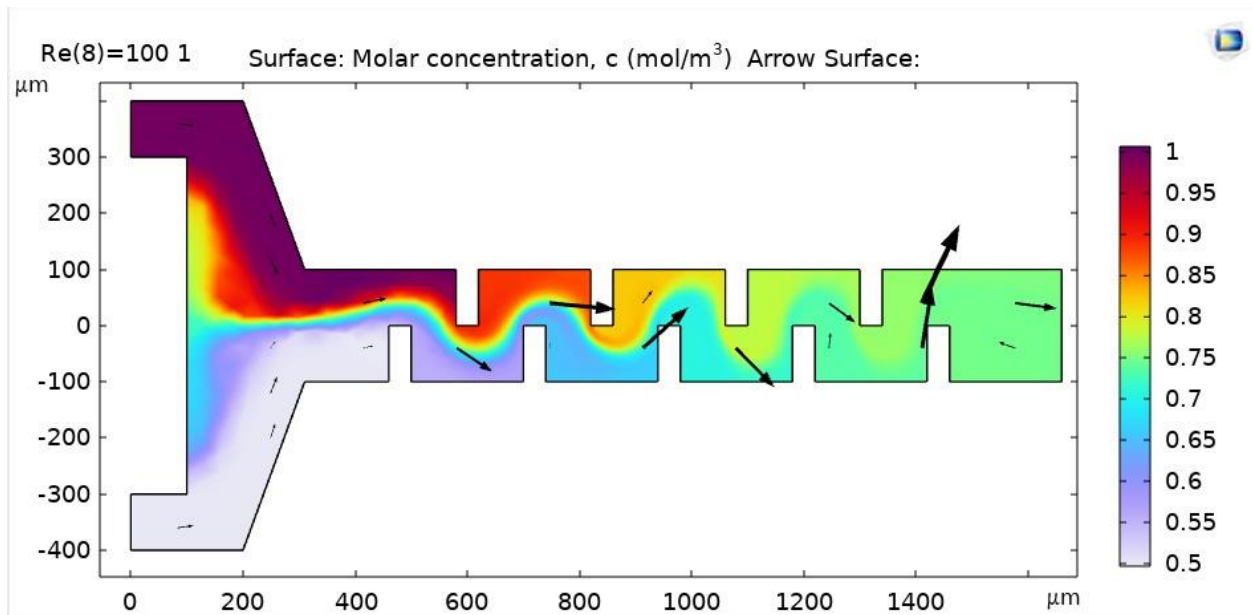
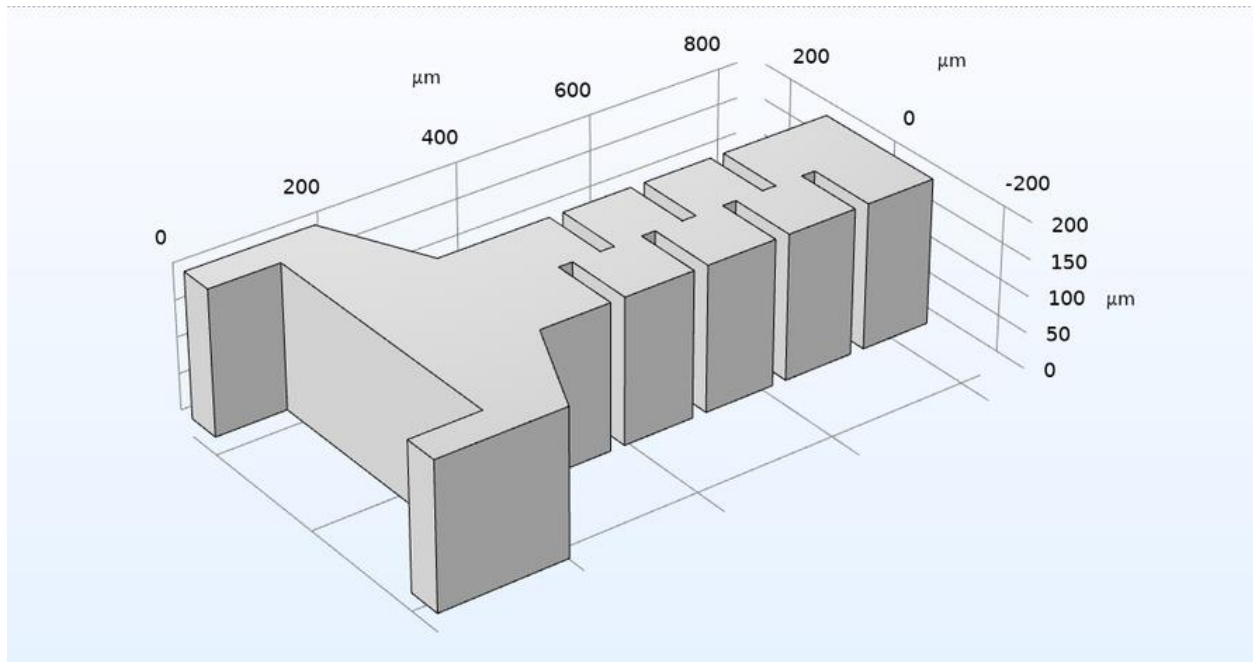
Micromixer Parameters

We are using 4 different Micromixer designs; their parameters and fluid properties are given below:

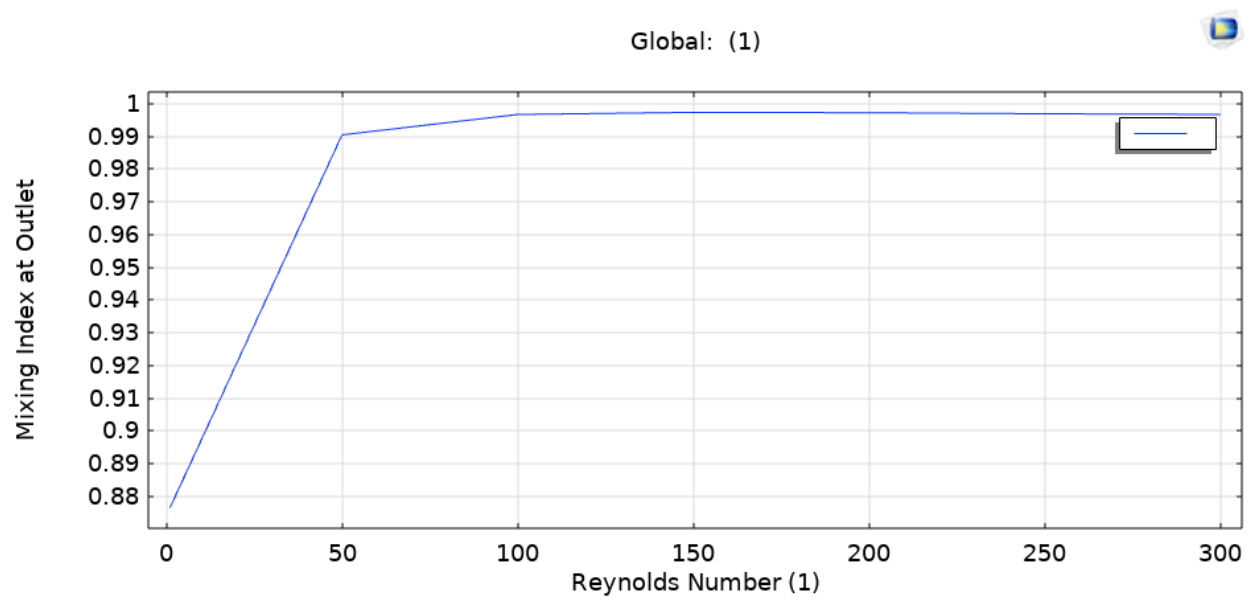
» Name	Expression	Value	Description
rho_w	1000 [kg/m ³]	1000 kg/m ³	Density of water
mu_w	0.001 [Pa·s]	0.001 Pa·s	Dynamic viscosity of water
L	200 [um]	2E-4 m	Channel Width
v_in1	0.2 [m/s]	0.2 m/s	Inlet 1 velocity
v_in2	0.2 [m/s]	0.2 m/s	Inlet 2 velocity
v_inavg	(v_in1+v_in2)/2	0.2 m/s	Inlet average velocity
D	10 ⁻⁹ [m ² /s]	1E-9 m ² /s	Diffusion coefficient
c_in1	1 [mol/m ³]	1 mol/m ³	Inflow 1 concentration
c_in2	0.5 [mol/m ³]	0.5 mol/m ³	Inflow 2 concentration
Re	(rho_w * v_inavg * L) / mu_w	40	Reynolds Number
Pe	(v_inavg * L) / D	40000	Peclet Number

6 Results and Discussion

6.1 Design 1 – Rectangular Ridges

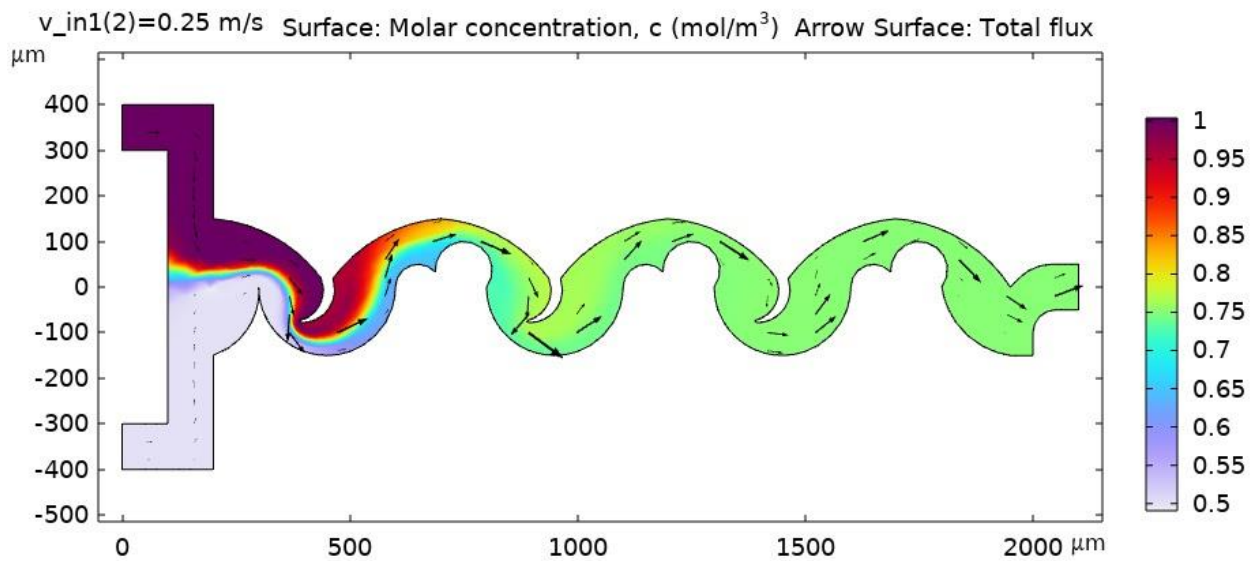
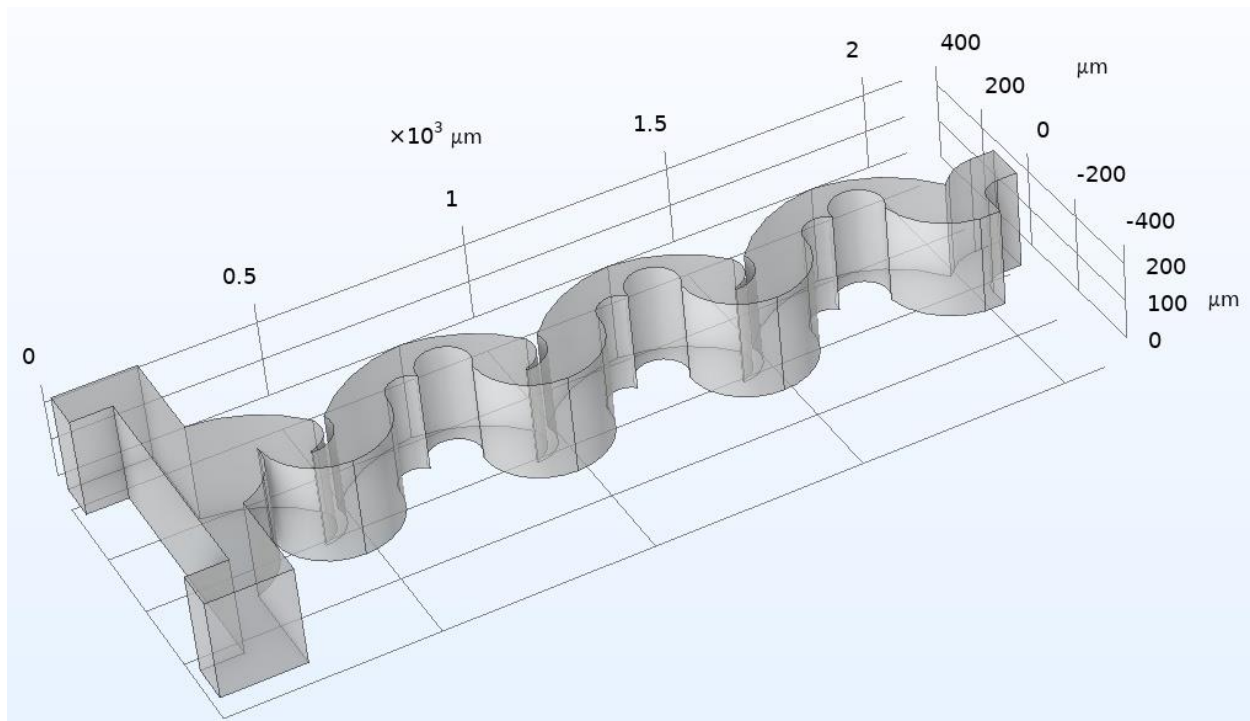


- Repeated rectangular incisions, causing sudden expansion and contraction.
- Interpretation: At low $Re \rightarrow$ weak disturbance. As Re increases $\rightarrow$ stronger recirculation inside cavities. Flow separation increases. Interface stretching increases. At higher Re , strong local vortices, hence better mixing.

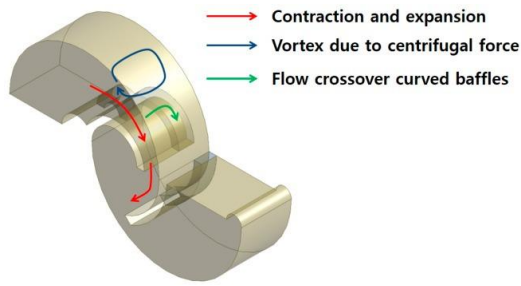


This is a recirculation-enhanced mixer with good inertial enhancement and physically consistent increasing trend.

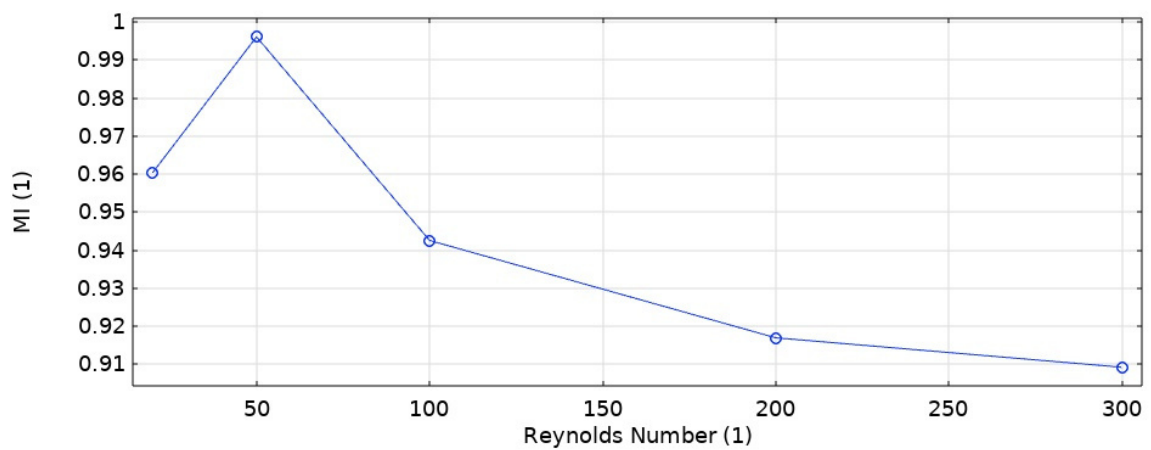
6.2 Design 2 – Curly Baffles



- Smooth circular curves, implies curvature induced centrifugal forces.

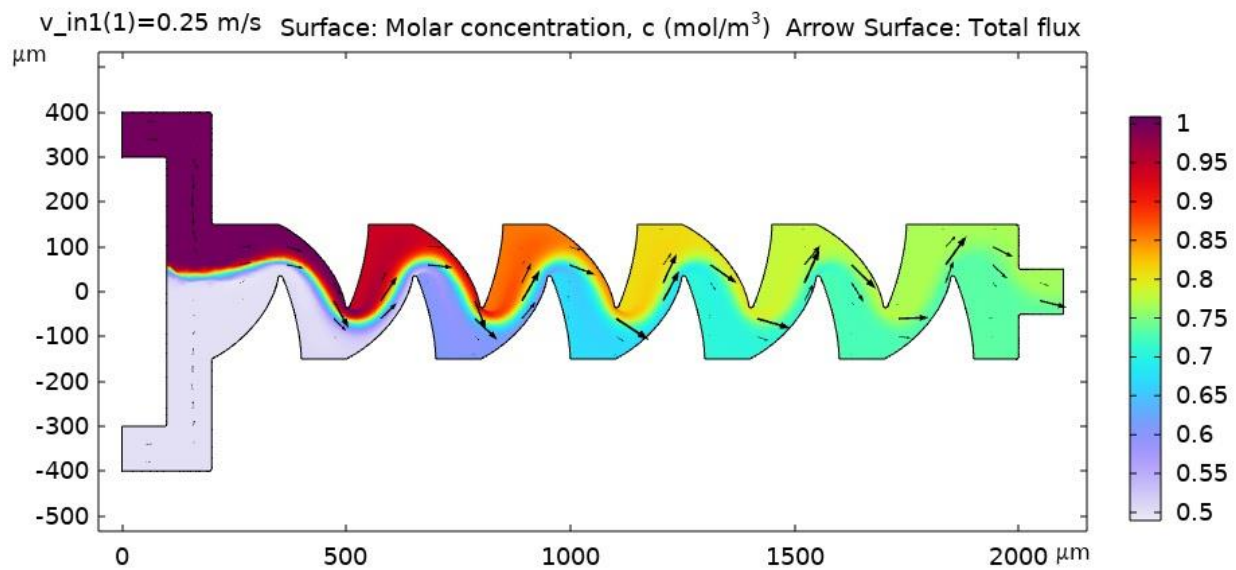
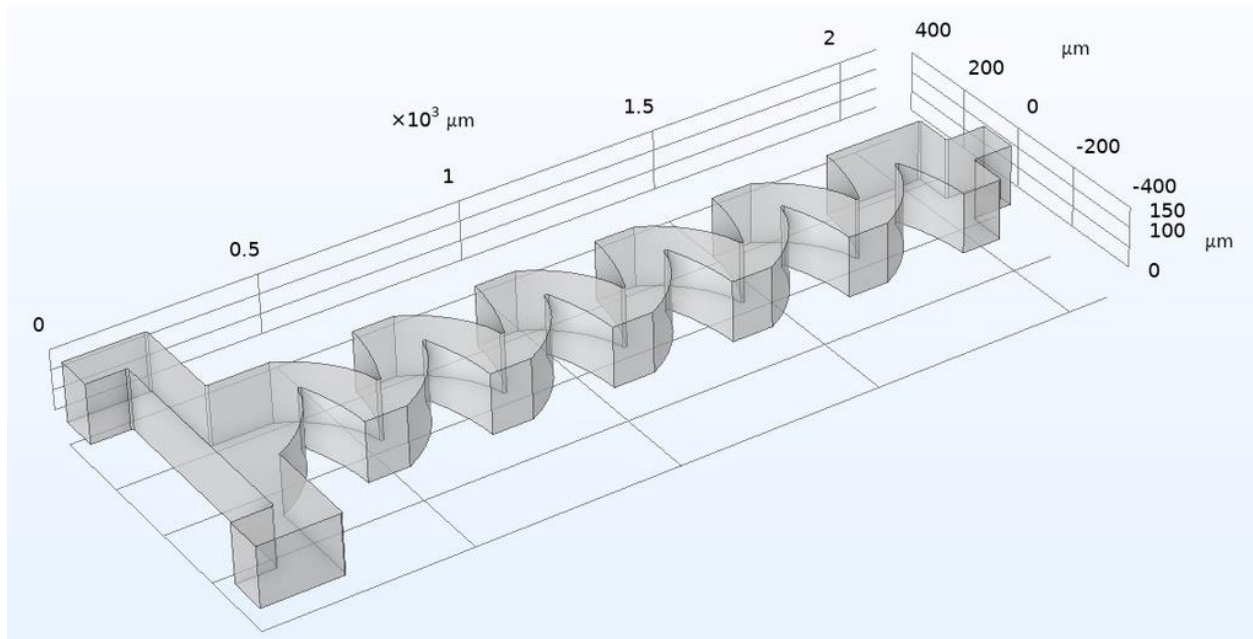


- Interpretation: It achieves the maximum efficiency in the shortest length of around 700 - 900 μm . But the MI trend peaks at $\text{Re} \approx 50$, then decreases strongly. This occurs because at higher Re the flow may no longer be steady and the diffusion may be insufficient as there would be no vortex compensation.

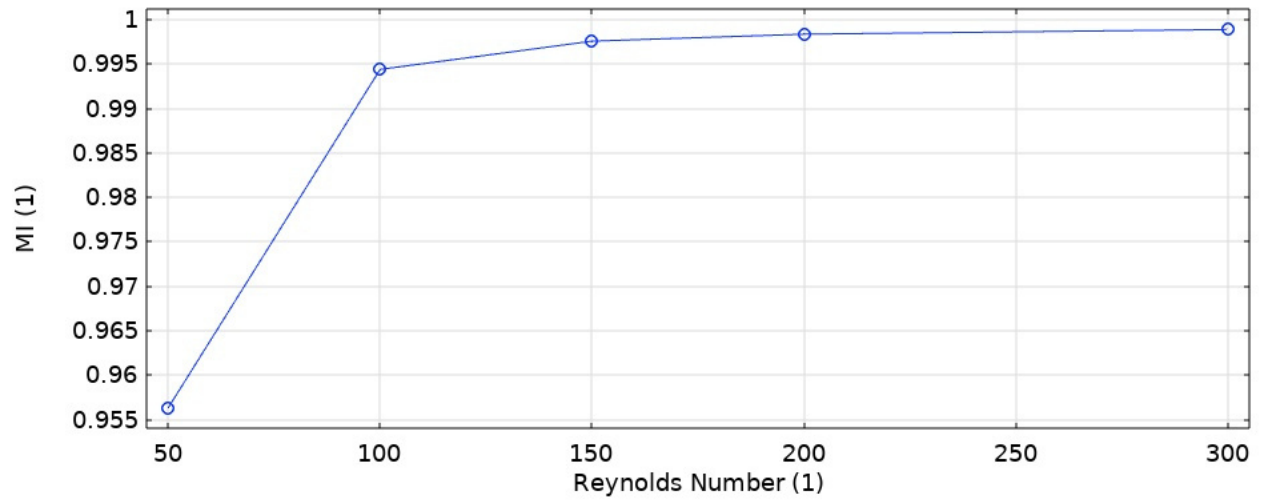


Overall, this is a very good mixer structure at lower Re but considering wide range of Re it is mixing tendency gets limited.

6.3 Design 3 – Fin Shaped

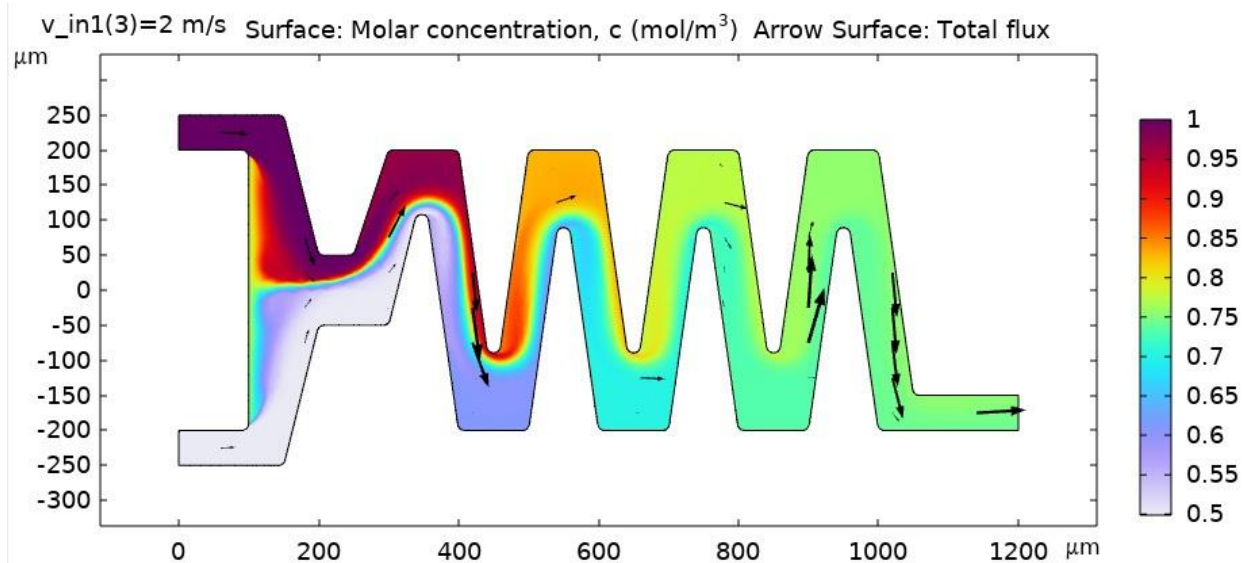
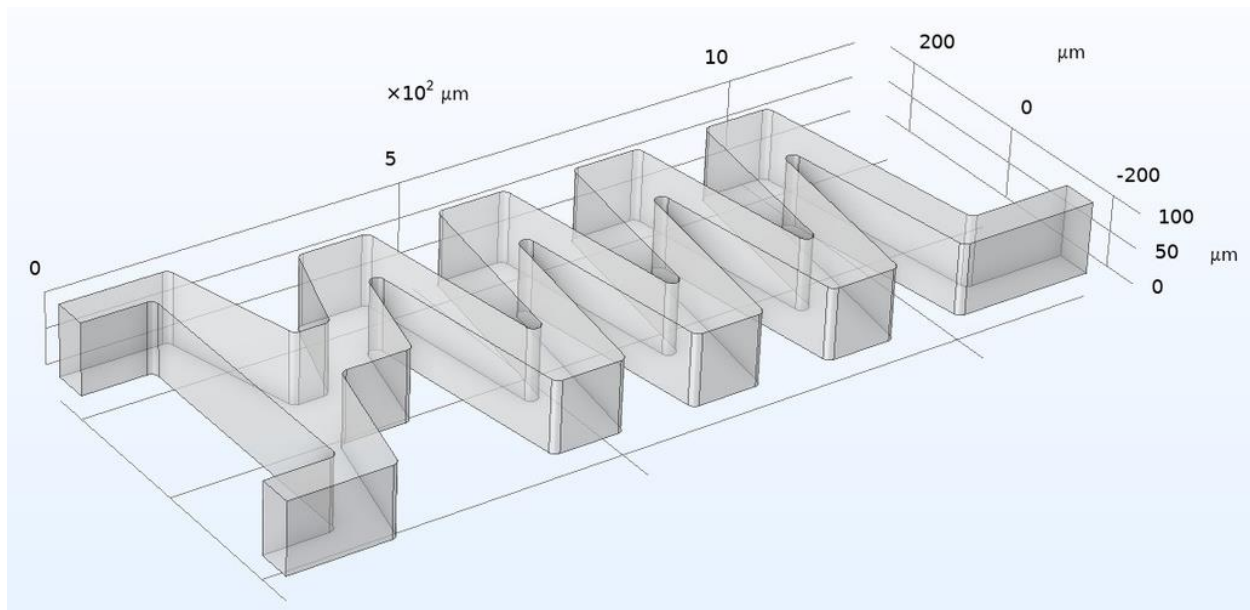


- Serpentine curved asymmetric cross section with repeated downward expansion.
- Interpretation - Diffusion dominates strongly at low Re already. At higher Re, Slightly stronger curvature effect is visible but mixing was already nearly complete.



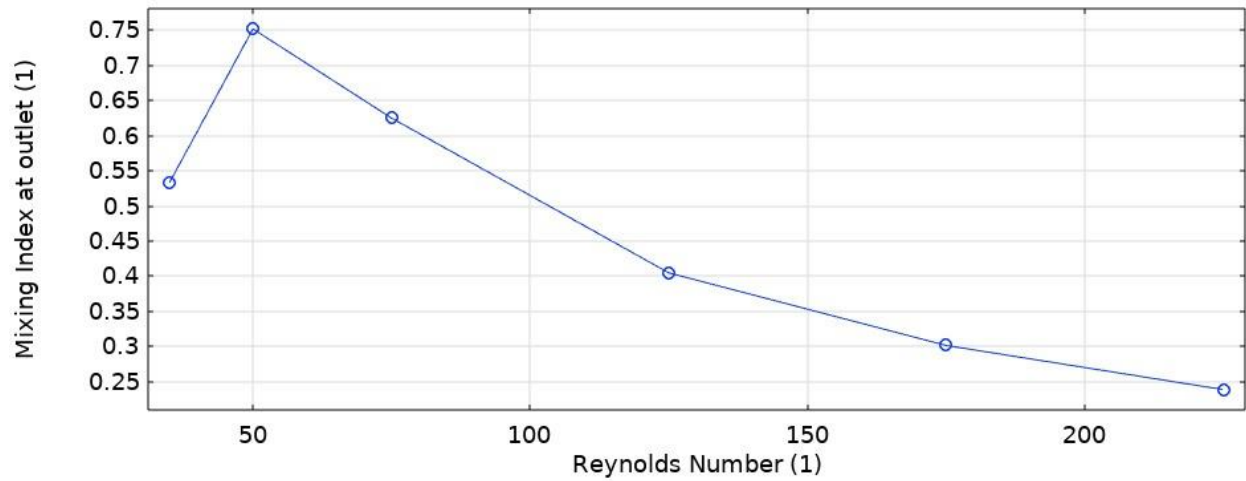
This is a very effective and flexible, diffusion-dominated mixer.

6.4 Design 4 – River Shaped



- Sharp V shaped turns with large directional deflection, local acceleration and deceleration zones.
- Interpretation - It exhibits an optimal mixing performance at Re around 50, where convective stretching and molecular diffusion act synergistically. Beyond this point,

increasing Re causes progressive decline in mixing efficiency due to absence of strong vortex-induced secondary flows.



This is a stretching-dominated but diffusion-limited micromixer. Thus not a strong inertial micromixer.

7 Conclusions

We can see that the four micromixer designs have vastly different MI vs Re trends. This shows that mixing is highly sensitive to obstacle geometry, since even slight changes can cause large shifts in diffusion, flow separation and vortex generation effects.

The most interesting design among the four is the curly baffles, with centrifugal forces present due to its circular curves:

- At very low Re, slow flow and very small centrifugal force, implying a low diffusion and vortex generation. This leads to low mixing rates.
- At moderate Re (around 50), ideal velocity range for best residence time and best vortex generation due to centrifugal forces. This results in the peak MI value.
- At higher Re, velocity becomes large leading to reduction in residence time and thereby less diffusion accompanied by reduction in formation of vortices, hence the decreasing MI with Re.
- Due to these factors and early recirculation by centrifugal forces, Curly baffles has the shortest mixing length.

Best micromixer design – Curly Baffles

In general, the design comparison on the basis of Re regime of inertial microfluids as concluded from the graphs are:

Rectangular Ridges > Fin Shaped > Curly Baffles > River Shaped

8 Future Scope

- **Transition from Diffusion-Dominated to Inertial Microfluidics.** Development of high-throughput micromixers capable of fast processing will take place.
- It will help in **Lab-on-Chip and Point-of-Care Diagnostics.** Help in the areas of Rapid disease detection (PCR chips), Blood analysis systems, etc.
- It will also help in **industrial development** like Microreactors for Chemical and Pharmaceutical Manufacturing. Industry is shifting towards continuous microreactors instead of batch reactors.
- Good scope for other **biomedical advancement** like cancer cell isolation, stem cell processing, etc.

9 References

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